

1. Introduction

Purpose of the Project

The purpose of this project is to analyze historical stock market data of selected technology companies using Python. The project focuses on understanding stock price movements, daily returns, risk, volatility, and the relationship between different stocks through statistical analysis and visualization.

Companies Selected

The following three technology companies were selected:

- **Apple Inc. (AAPL)**
- **Amazon.com Inc. (AMZN)**
- **Netflix Inc. (NFLX)**

Reason for Selecting These Companies

These companies were selected because they are major technology companies with actively traded stocks. Comparing them provides an opportunity to understand how different technology stocks performed during the same period and how their prices and returns were related to each other.

2. Methodology

Data Source

Historical daily stock market data was collected using the **yFinance Python library**. Data for the period **January 2024 to December 2024** was collected for AAPL, AMZN, and NFLX.

Each company had **252 daily records** after data collection.

Python Libraries Used

The following Python libraries were used:

- **Pandas** – Data manipulation and statistical analysis
- **NumPy** – Numerical calculations
- **Matplotlib** – Creating graphs and visualizations
- **Seaborn** – Creating the correlation heatmap
- **yFinance** – Collecting historical stock market data

Analysis Techniques Applied

The following analyses were performed:

1. Missing-value checking and data preprocessing
2. Descriptive statistical analysis
3. Closing-price trend analysis
4. Trading-volume analysis
5. Daily return calculation
6. Volatility analysis using standard deviation
7. Pearson correlation analysis
8. Correlation heatmap visualization

3. Results and Findings

A. Descriptive Statistics

The following measures were calculated for each company:

- Mean closing price
- Median closing price
- Minimum closing price
- Maximum closing price
- Standard deviation

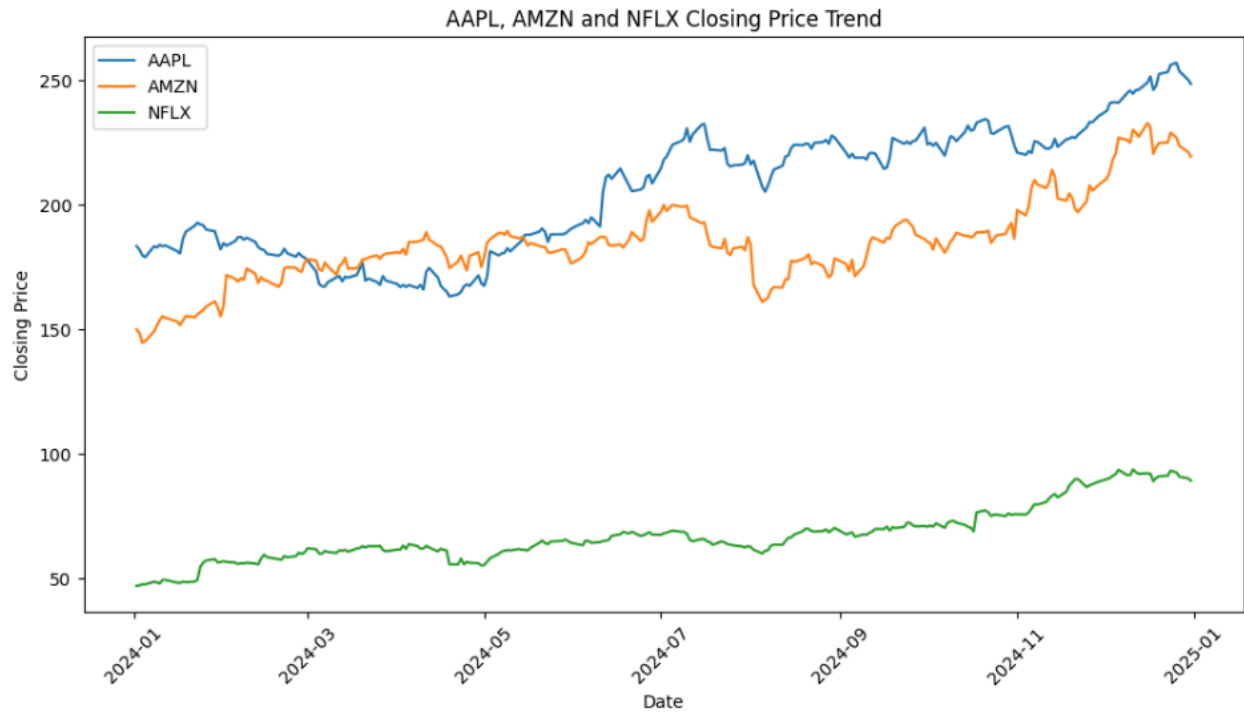
For example:

...	Company	Mean	Median	Minimum	Maximum	Standard Deviation
0	AAPL	205.280350	212.241814	163.220642	257.153809	25.530140
1	AMZN	184.628691	183.430000	144.570007	232.929993	17.427537
2	NFLX	67.150024	64.784000	46.849998	93.655998	11.023968

B. Stock Price Analysis

The closing-price line chart showed that all three stocks experienced fluctuations during the selected period. Despite temporary increases and decreases, the three stocks showed an **overall increasing trend**.

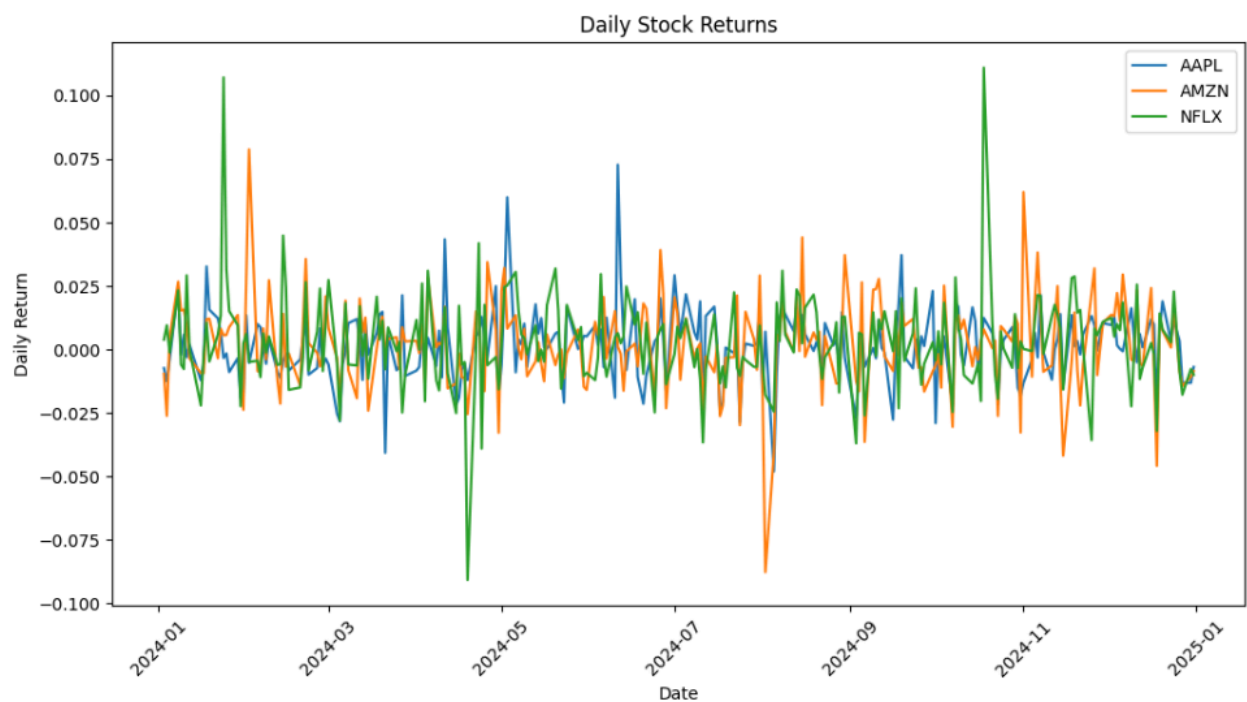
The stock prices changed throughout the year, with several periods of growth and temporary declines.



C. Daily Return Analysis

Daily returns were calculated using the formula:

$$\text{Daily Return} = \frac{\text{Today's Closing Price} - \text{Previous Day's Closing Price}}{\text{Previous Day's Closing Price}}$$

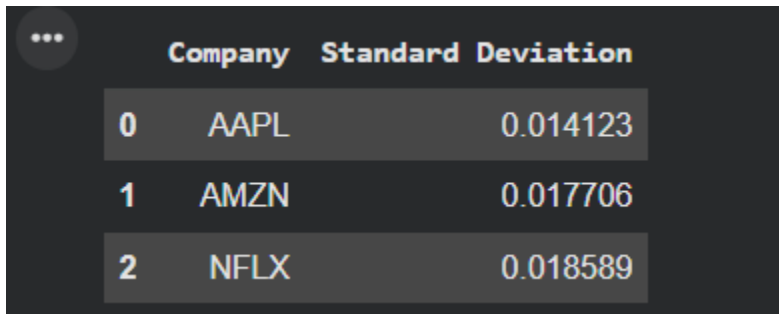


The daily return graph showed frequent fluctuations around zero. This indicates that stock prices changed both positively and negatively from one trading day to another.

The largest positive and negative movements were particularly noticeable for NFLX.

D. Volatility Analysis

Standard deviation of daily returns was used to measure volatility.



A screenshot of a table with a dark background. The table has two columns: 'Company' and 'Standard Deviation'. There are three rows of data. The first row is for AAPL with a standard deviation of 0.014123. The second row is for AMZN with a standard deviation of 0.017706. The third row is for NFLX with a standard deviation of 0.018589. The table is indexed from 0 to 2.

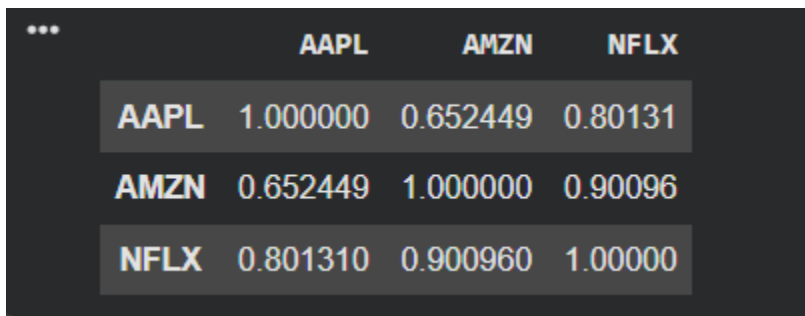
	Company	Standard Deviation
0	AAPL	0.014123
1	AMZN	0.017706
2	NFLX	0.018589

The results show that:

- **NFLX** had the highest volatility (**0.018589**).
 - **AAPL** had the lowest volatility (**0.014123**).
 - Therefore, NFLX experienced the greatest variation in its daily returns.
 - AAPL was the most stable among the three stocks.
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E. Correlation Analysis

Pearson correlation was used to measure the relationship between the closing prices of the three companies. Your correlation results were:



A screenshot of a correlation matrix table with a dark background. The table has four columns: the first column lists the companies (AAPL, AMZN, NFLX), and the next three columns show the Pearson correlation coefficients between each pair of companies. The diagonal elements are all 1.000000. The correlation between AAPL and AMZN is 0.652449, between AAPL and NFLX is 0.801310, and between AMZN and NFLX is 0.900960.

	AAPL	AMZN	NFLX
AAPL	1.000000	0.652449	0.801310
AMZN	0.652449	1.000000	0.900960
NFLX	0.801310	0.900960	1.000000

The results show that:

- **AMZN and NFLX** had the strongest relationship with a correlation of **0.90**.
- **AAPL and AMZN** had the weakest relationship with a correlation of **0.65**.
- All three relationships were positive, meaning the stocks generally tended to move in the same direction.

4. Conclusion

1. Which company performed best during the selected period?

Based on the overall closing-price movement during the selected period, **NFLX showed the strongest overall price growth** among the three selected stocks.

2. Which stock was the riskiest?

NFLX was the riskiest stock because it had the highest standard deviation of daily returns (**0.018589**), indicating greater price fluctuations.

3. Which stock was the most stable?

AAPL was the most stable stock because it had the lowest standard deviation of daily returns (**0.014123**).

4. What relationship was found between different companies?

A positive relationship was found between all three companies. The **strongest relationship was between AMZN and NFLX (0.90)**, while the **weakest relationship was between AAPL and AMZN (0.65)**. This indicates that the selected technology stocks generally moved in the same direction during the selected period.

Overall Conclusion

The analysis showed that all three technology stocks experienced fluctuations but generally followed an increasing trend during 2024. NFLX showed the strongest overall performance but also had the highest volatility, making it the riskiest stock. AAPL was the most stable stock due to its lower volatility. The correlation analysis showed positive relationships among all three companies, with the strongest relationship observed between AMZN and NFLX.

